

Abubakar Khawaja

Senior Backend Engineer | Python | Django | Node.js | PostgreSQL | AWS

abubakar-dev@hotmail.com • +92-334-9858841

[linkedin.com/in/abubakar-khawaja-008483183](https://www.linkedin.com/in/abubakar-khawaja-008483183) • github.com/Abuubkar • abuubkar.github.io

PROFESSIONAL SUMMARY

Backend Engineer with 5+ years building and running production services in Python and Django, with Django REST Framework, PostgreSQL, Redis, and Celery. I design REST APIs, tune database performance, and move heavy work onto asynchronous pipelines that stay stable under load. I have run services on AWS using ECS, EKS, and Lambda, with PostgreSQL on RDS and Kubernetes for orchestration, and I build Node.js APIs with NestJS and TypeScript alongside the Python work. I am comfortable owning data modelling, API security, and background job throughput end to end.

EXPERIENCE

Senior Python Developer | Arbisoft

2021 – 2026

- Architected asynchronous processing pipelines using Celery and Redis for bulk document generation, scheduled jobs, and reporting systems managing 1,000+ assets.
- Designed and maintained scalable Django applications serving production workloads across multiple business domains.
- Designed secure REST APIs with Django REST Framework, implementing JWT authentication, authorization, and third-party integrations.
- Cut API response times by 30-70% across core endpoints by resolving N+1 queries, optimizing ORM access patterns, and implementing strategic PostgreSQL indexing.
- Diagnosed a Celery task queue backlog and Redis memory bloat, then moved message routing onto RabbitMQ to stabilize background job throughput.
- Deployed and tuned services on AWS ECS/EKS and Lambda to support scaling, build performance, and deployment reliability.
- Built and maintained CI/CD pipelines using GitLab CI and GitHub Actions while containerizing applications with Docker for reliable deployments.
- Improved code quality through automated testing, code reviews, and development best practices, increasing automated test coverage beyond 60%.
- Owned Sentry monitoring to cut production error volume: error triage with session replay, release and performance tracking, Slack and GitHub integration, and PII-safe payload encryption that keeps sensitive data out of captured events.
- Built Scrapy spiders to collect job listings from public sites, using tuned download delays and user-agent rotation to stay within source rate limits.
- Maintained services across several repositories in Open edX, a learning platform built from microservices, and shipped bug fixes and version upgrades.

SELECTED PROJECTS

Telehealth Platform · [Django](#) + [PostgreSQL](#) + [Kubernetes](#)

- Architected and maintained a secure, HIPAA-compliant telehealth platform using Django, PostgreSQL on AWS RDS, and Kubernetes for container orchestration.
- Optimized patient-provider matching workflows through SQL query tuning and Redis caching, collaborating with data engineering to streamline ETL pipelines for real-time data processing.
- Integrated HL7 ADT (admit, discharge, transfer) message processing so patient events sync between clinical systems as they happen, with HIPAA-compliant handling of PHI in transit.

FixAlert, Maintenance Platform · [Django](#) + [PostgreSQL](#)

- Implemented Celery-based background processing, Stripe payment integration, and scalable reporting services.
- Optimized backend performance by eliminating N+1 queries and improving database access patterns.
- Made Stripe payment requests retry-safe using Stripe's idempotency key header, so a retried call could not create a duplicate charge.

Landit, Career Platform · [Django](#) + [React](#)

- Owned the design and delivery of a complex analytics dashboard in Google Looker, giving stakeholders real-time visibility into key platform metrics.
- Built transactional email workflows with SendGrid and optimized backend services.

Foodio, Food Delivery Platform · [NestJS](#) + [TypeScript](#) + [PostgreSQL/PostGIS](#) · [Personal Project](#)

- Architected a Node.js food-delivery API on NestJS and Express with feature-first modules for identity, ordering, billing, payments, and notifications, on PostgreSQL with PostGIS for geo-based restaurant discovery via Drizzle ORM.
- Designed order state as an append-only event log with derived status, and made order placement idempotent via a client-generated Idempotency-Key so a retried request could never create a duplicate order.
- Shared a Zod contract package between the backend and the React Native app for runtime-validated DTOs, and set up GitHub Actions CI running typecheck, lint, Vitest unit tests, and Testcontainers integration tests against real PostGIS, with Docker Compose and a production Dockerfile.

CORE SKILLS

Languages: Python, SQL, JavaScript, TypeScript

Backend: Python (Django, Django REST Framework, FastAPI, Flask), Node.js (NestJS, Express), REST API design, GraphQL, webhooks, JWT & OAuth2 authentication, Celery & RabbitMQ (async jobs), WebSockets, Scrappy

Databases: PostgreSQL, MongoDB, DynamoDB, Redis, Elasticsearch, MySQL, PostGIS, Database Indexing, Query Optimization, Caching

Cloud & DevOps: AWS (ECS, EKS, Lambda, S3, RDS, Cognito, API Gateway), Docker, Kubernetes, GitHub Actions, GitLab CI/CD, Datadog, Sentry, Nginx, Linux, Git

Testing: Pytest, Django Test Framework, Vitest, Testcontainers, Unit & Integration Testing, API Testing

Software Engineering: Object-Oriented Programming (OOP), Design Patterns, Microservices, System Design, Performance Optimization, Code Review, Agile/Scrum

Backend Integration: REST APIs, GraphQL, API Design Collaboration

Integrations: Stripe (payments), SendGrid & Resend (email), Firebase Auth, Google Maps, PostHog & Google Analytics, Google Looker (BI), JWT/OAuth2

AI/ML: LLM Application Architecture, AI Agents, Tool/Function Calling, Structured Outputs (Zod, Pydantic), Model Context Protocol (MCP), RAG, Embeddings & Vector Databases (pgvector), Prompt & Context Engineering, OpenAI / OpenRouter / Ollama (local models), Vercel AI SDK

AI Development: ChatGPT, GitHub Copilot, Cursor, Claude Code, Codex, Prompt Engineering, AI-Assisted Development

EDUCATION

B.S. Software Engineering Punjab University College of Information & Technology (PUCIT) · GPA: 3.08

CERTIFICATIONS

- The AI Engineer Path & Prompt Engineering for Web Developers — Scrimba (July 2026)